ORAL DEFENSE · GROUP 6

SIDEFLIX

Which independent films should we feature this week?

Matis Douay · Jérémy Caillaud · Rayan Faher Bonne

Data Engineering & Cloud Infrastructure · Albert School · 6 November 2026

Daily batch · 06:00 Paris | Snowflake · AWS Paris | dbt | Airflow in Docker | Terraform

Speaker: Matis

Sideflix: a fictional indie-cinema streamer

FICTIONAL COMPANY · REAL-WORLD BENCHMARK: NETFLIX

10,000	€7.99	≈ €80k	100	6
subscribers, film fans	per month, ad-free	revenue per month (€0.96m a year)	titles in the scenario catalogue	employees

Revenue & customers

- Subscriptions only: 10,000 × €7.99 ≈ €80k a month
- Film fans who want an ad-free niche catalogue

Operations & stakeholders

- 6 employees; featured titles reviewed every week
- Subscribers, catalogue manager, independent producers, management

Problem & constraints

- 100 titles compete for weekly featuring
- Figures ready by 07:00 Paris daily
- Warehouse budget: 10 credits/month

Possible exit: a sale to Netflix, as a scenario only. All figures are explicit assumptions; the prototype runs on 6 real films.

Evidence: README.md, Step 1 (scenario assumptions) · evidence/README.md, C1

Speaker: Matis

One user, one weekly decision

THE USER

Catalogue manager

Which titles should we feature this week?

- **06:00** — Daily run starts (Paris time)
- **< 07:00** — Target: figures ready (a run takes about 1 min)
- **Weekly** — Featuring decision on the last 7 days

Metric

- Valid sessions = distinct session_id after exclusions
- Minutes watched = sum of watched seconds ÷ 60

Grain & cadence

- Title × UTC day of session start; ranking over 7 days by minutes, then sessions
- Daily run keeps the window current before the weekly review

Exclusions

- Quarantined with reason: missing ID, unknown title, bad start time, duration ≤ 0 or > runtime
- Duplicates removed · an unexpected device value fails a test

Acceptance test

- 1 Nov 2026: 13 valid sessions, 177.4 minutes
- Equal to an independent calculation from the input files

Evidence: README.md, Step 1 · source/v2/expected_daily_usage.csv · evidence/README.md, C2

Speaker: Matis

Five real sources describe the films

REAL AND PUBLIC · THEY DESCRIBE THE FILMS AND THE MARKET, NOT SIDEFLIX VIEWING

IMDb

Identity & runtime

Shared key: tconst. Sessions longer than the film are quarantined.

Snapshot · 4 Nov 2026

FILE → COPY

Wikidata

Film ↔ Wikipedia

One QID per film, with its IMDb ID and article title.

Live call · 4 Nov snapshot

API → VARIANT

Wikimedia

Popularity now

42 daily page-view values for the six films.

26 Oct – 1 Nov 2026

API → VARIANT

MovieLens

Perceived quality

74 ratings: average and count per film, always shown together.

Our ratings · 2015–2018

FILE → COPY

Netflix Top 10

Real-world success

Weekly global ranks. Only confirmed match: Sicario, rank 5.

Week ending 9 Mar 2025

FILE → COPY

INTERNAL AND PRIVATE · SIDEFLIX'S OWN DATA

Catalogue · local PostgreSQL

Six films with their IMDb runtime · simulated on-premises database in Docker · watermark + MERGE

Viewing sessions

SIMULATED

Viewing data is private, so it is simulated: 32 rows · 6 films
30 Oct – 1 Nov 2026

Evidence: source/sources_manifest.csv (URL, SHA-256, period, terms) · docs/sources_and_catalogue.md · evidence/01_sources

Speaker: Matis

Simulated viewing meets real context

Simulated sessions

- 1 session = 1 viewing
- 32 raw rows · 6 films · 3 days
- 30 valid + 1 quarantined (S31, −60 s) + 1 duplicate (S4)

Silver · validate

- The IMDb catalogue validates each session: known title, 0 < duration ≤ runtime
- Duplicates collapse to one session_id; invalid rows keep their reason

Gold · rank + context

- Ranking from Sideflix viewing only: minutes, then sessions
- MovieLens, Wikipedia and Netflix shown next to it, never blended into a score

Real sources join twice: in Silver (IMDb validates) and in Gold (context next to the rank).

READING 1 NOVEMBER 2026

14	− S31	13	177.4	3	7
raw rows that day	quarantined (−60 s)	valid sessions	minutes watched	max for one film (Frances Ha)	Whiplash sessions over the 3 days

Evidence: source/v2/sessions/ · source/v2/expected_daily_usage.csv · evidence/04_dbt_and_quality/quality_report.md

Speaker: Matis

Three source shapes, three ingestion modes

PostgreSQL catalogue

rows that change

updated_at watermark + MERGE

TD 2, part B

NEED

Load only rows changed since the last run.

CONTROLS

- MERGE on content_id; each run logged in LOAD_RUNS
- A replay updates nothing

WHY NOT THE OTHERS

Full reload: simple and catches deletes, but resends the whole table daily. CDC: needs log access.

Files

IMDb, MovieLens, Netflix, sessions

PUT to stage + COPY INTO

TD 2, part A

NEED

Load each snapshot batch once.

CONTROLS

- Known batch (content hash) is skipped
- COPY aborts on error; row counts checked

WHY NOT THE OTHERS

Snowpipe: made for files that arrive continuously; ours arrive once a day.

Public APIs

Wikidata, Wikimedia

requests → JSON → VARIANT

resource pack 05

NEED

Keep raw JSON; replay offline.

CONTROLS

- 30 s timeout, user agent, retry on 429
- Responses saved; typed in Silver

WHY NOT THE OTHERS

Flattening in Python: Bronze would keep no raw JSON to re-type or replay.

Everything lands in Bronze with load time and batch ID; selected file rows and full API responses land untyped.

Evidence: ingestion/incremental_catalog.py · ingestion/load_sessions.py · evidence/02_ingestion/ingestion_report.md

Speaker: Matis

Bad rows stop early; replays add nothing

1 check_sources

Each file and API batch complete with its load metadata; sessions present for the date

→

2 Silver (dbt)

Safe typing (TRY_), quarantine with a reason, one row per session_id

→

3 dbt tests

21 tests:
15 generic + 6 singular,
keys, rules, Gold = Silver

→

4 Publish

MERGE keyed on the date, only after a fully green build

A failed test skips the Gold tables downstream of it, and any failure blocks publication: a bad row never reaches the published figure.

RECONCILIATION TEST

32 = 30+1+1

raw rows = valid + quarantined + duplicate

BUILD SUMMARY

33 / 33

dbt operations: 11 models + 21 tests + 1 audit hook

FRESHNESS

2 / 2

source freshness checks, read from the ingestion journal

REPLAY TEST

0

new rows when the same date is replayed twice

Evidence: evidence/04_dbt_and_quality/quality_report.md · evidence/06_incidents/incident_9_4_replay.json · dbt/sideflix_dbt/

Speaker: Jérémy

Proportionate tools for one daily batch

TOOL	ROLE IN SIDEFLIX	ALTERNATIVE · TRADE-OFF
Snowflake AWS eu-west-3 (Paris)	Stores and computes Bronze, Silver, Gold on an X-Small warehouse that suspends after 60 s idle.	BigQuery fits too; our course account already gave us stage/COPY, VARIANT, key-pair auth and dbt.
dbt 9 Silver views · 2 Gold tables	Versioned SQL with 21 tests, generated docs and lineage.	Stored procedures: same SQL, but no ref lineage, tests or docs.
Airflow 3.3 in Docker DockerOperator	06:00 Paris, 2 retries, Slack alert, replay by date; every step in a container.	Snowflake Tasks: fewer parts, but cannot reach our local PostgreSQL; API calls need an external access integration.
Terraform warehouse as code	SIDEFLIX_WH declared in code; state in a private, versioned, locked GCS bucket.	S3 would keep one cloud, but no AWS access was set up on the state holder's workstation; GCS holds only the state (disclosed).

Batch, not streaming

A weekly decision needs one daily batch. Kafka, CDC, Snowpipe or Iceberg would add operations without value here.

One run takes about a minute (59 s in the final rehearsal), well before 07:00.

PROOF · REGION QUERY

```
SELECT CURRENT_REGION();
```

AWS_EU_WEST_3

Every run traverses Snowflake on AWS (Paris).

Evidence: docs/architecture.md · evidence/03_cloud_and_iac (region, Terraform) · terraform/ · airflow/dags/sideflix_daily.py

Speaker: Jérémy

About 1.5 credits a month, capped at 10

1.44

credits measured

3 Nov – 4 Nov 18:10 UTC: development, tests and incident runs

≈ 1.5

credits per month, projected

30 runs × 131 s × 1.35 credits/h (X-Small Gen2), one run a day: a projection, not an invoice

10

credits per month, monitor limit

Notify at 50% and 80%; suspend at 100%, once running queries finish

Least privilege

- Pipeline role SIDEFLIX_RUNNER: no DELETE on Bronze
- Service user SIDEFLIX_PIPELINE: key pair, no password
- ANALYST: SELECT on Gold only · 11 checks: 3 allowed, 8 denied

Secrets & data

- RSA key folder mounted read-only at run time
- .env, keys, webhook and Terraform state git-ignored
- Pseudonymous viewer_id stays in Bronze/Silver; the analyst reads aggregated Gold

Shutdown & clean-up

- Documented: pause the DAG, docker compose down, suspend the warehouse
- Clean-up only after grading, never the course database

Evidence: evidence/07_security_and_costs/security_costs_report.md · access_tests.txt · governance/README.md

Speaker: Rayan

The executed architecture, end to end

SOURCES AND LOCAL INGESTION JOBS

File snapshots
IMDb · MovieLens · Netflix Top 10
+ 32 simulated session records

PostgreSQL source-db
Catalogue of the 6 films
local Docker Compose container

Public APIs
Wikidata metadata
Wikimedia page views, live calls

File ingestion
PUT · stage · COPY INTO · TD 2A

Incremental catalogue
watermark + MERGE · TD 2B

API ingestion
requests → JSON → VARIANT · pack 05

SNOWFLAKE · AWS eu-west-3 (Paris) · SIDEFLIX_WH, X-Small

BRONZE
Raw values + batch,
file and load time

SILVER · dbt
9 views: typing,
dedup, quarantine

GOLD · dbt
2 tables: daily usage
+ 7-day ranking

PUBLISH · CONSUME
LAB.DAILY_USAGE (MERGE)
Snowsight reads Gold

Docker + Airflow · local
06:00 Paris · 4 tasks · 2 retries
Slack alert after the last retry

Terraform
SIDEFLIX_WH declared in code
locked GCS backend, state only

Quality · access · cost
21 dbt tests + freshness gate
RUNNER / ANALYST · 10-credit cap

Evidence: docs/architecture.png (original in the appendix) · docs/architecture.md · evidence/README.md

Speaker: Rayan

Whiplash leads the 7-day ranking

GOLD RANKING · 7 DAYS

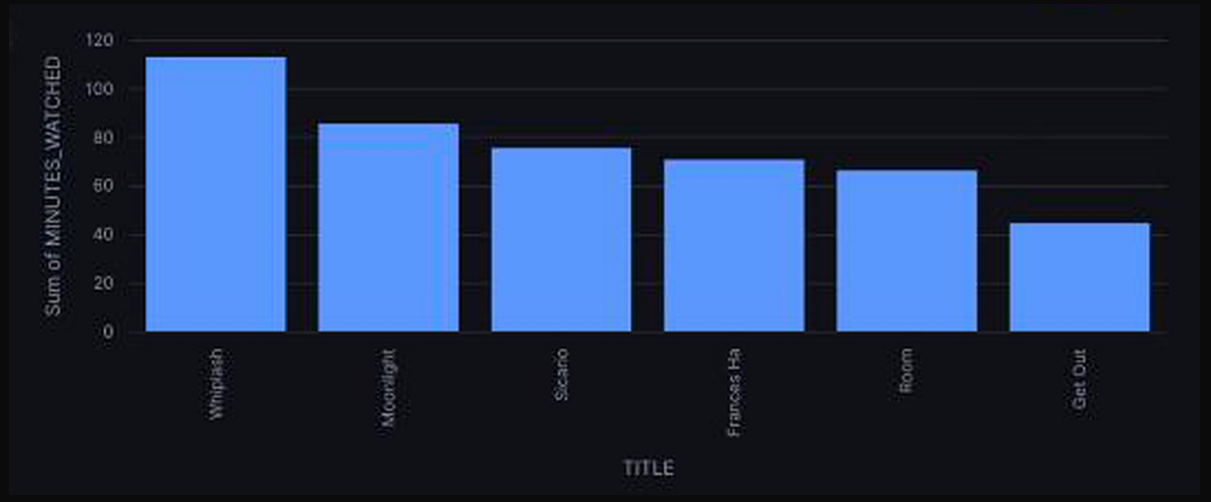
SIMULATED VIEWING

1	**Whiplash**	7 sessions	112.7
2	**Moonlight**	6 sessions	85.3
3	**Sicario**	4 sessions	75.3
4	**Frances Ha**	5 sessions	70.5
5	**Room**	4 sessions	66.1
6	**Get Out**	4 sessions	44.4

Window 26 Oct – 1 Nov 2026 · minutes, then sessions
30 valid sessions · 454.2 minutes

Which titles should we feature this week?

Source: SIDEFLIX.GOLD.GOLD_EDITORIAL_WEEKLY · read-only analyst role
Window: UTC days 26 Oct – 1 Nov 2026 · Gold built 4 Nov 2026, 19:48 UTC



Snowsight capture as recorded · readable values on the left

REAL CONTEXT, SHOWN NEXT TO THE RANK, NEVER BLENDED

Whiplash: MovieLens 4.04 (38 ratings, 2015–2018) · 15,551 Wikipedia views
Sicario: Netflix Top 10, English films: rank 5, week ending 9 Mar 2025 (only match)

Evidence: evidence/08_gold_report/ (snowsight_report.md, snowsight_sql_receipt.json) · docs/snowsight_gold_report.sql

Speakers: Jérémy · Matis · Rayan

Live run: one date in, one row out



LIVE · about 4 min

Rehearsed on 5 Nov 2026 with the delivered code:
4 green tasks, 95 Snowflake queries, 0 errors

Reproduce it (README)

- docker build: sideflix-ingest:1.0.0 and sideflix-dbt:1.0.0
- docker compose up → unpause → trigger with a date
- Job images: pinned python:3.12.7-slim, non-root, keys mounted read-only

0 — Before we start

Scheduled run for the previous day: no sessions → fails at check_sources, Slack alert

1 — Trigger {"date":"2026-11-01"}

Fresh run ID, no incident option: ingest → check → dbt build → publish

2 — While it runs: governance as SIDEFLIX_ANALYST

read Gold ✓ allowed · read Bronze ✗ denied · INSERT into Gold ✗ denied

3 — Airflow: 4 green tasks

freshness 2/2 · PASS=33 ERROR=0 · PUBLICATION_UNIQUE_DATE_PASS

4 — Snowsight as SIDEFLIX_RUNNER

One row for 2026-11-01: 13 sessions, 177.4 minutes, today's run ID

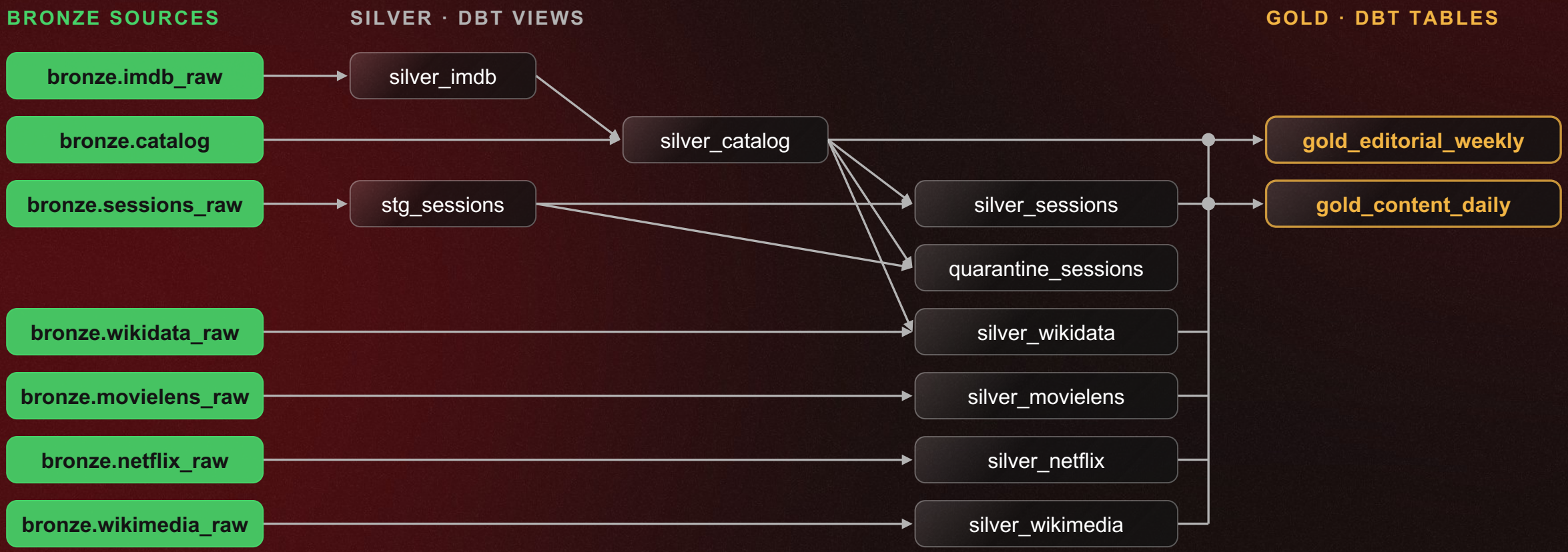
5 — Lineage and docs

docs/dbt/index.html: sessions_raw → stg_sessions → silver_sessions → gold_content_daily

Procedure: docs/live_demo_runbook.md · Fallback: evidence/10_final_check/final_check_verification.json · deliverables/Sideflix_pipeline.mov

Speaker: Matis

Every Gold figure traces back to Bronze



21 tests

15 generic on Silver: unique, not_null, accepted_values, relationships
6 singular: Gold = Silver · 32 = 30 + 1 + 1 · 0 < duration ≤ runtime · completeness · external grains · no inflation

Evidence: docs/dbt/manifest.json (drawn from it) · docs/dbt/index.html · docs/dbt_lineage.png (appendix)

Speakers: Matis · Jérémy

Two injected failures, two clean recoveries

LIVE · RUN FIRST · ≈ 1 MIN 30 S

9.2 · Crash after the write

{"date":"2026-11-01","incident":"crash_after_write"}

Attempt 1	publish logs MERGE_PUBLISHED, then a simulated crash
Retry	Airflow retries automatically after 20 s
Attempt 2	publishes again: PUBLICATION_UNIQUE_DATE_PASS
Result	still one row for 1 Nov: 13 sessions, 177.4 min

Why it holds: the publication is a MERGE keyed on the date. An INSERT would have counted the day twice.

LIVE IF TIME · ≈ 4–5 MIN WITH CLEAN-UP

9.1 · Invalid data

{"date":"2026-11-01","incident":"bad_device"}

Inject	isolated 32-row copy, one device = "invalid"
Detect	accepted_values fails ×3: PASS=28 ERROR=1 SKIP=4
Contain	publish upstream_failed · Slack alert · Gold unchanged
Recover	SYSADMIN deletes only the test batch → normal run 33/33

Why it holds: the failed test skipped both Gold tables and blocked publication; the invalid row reached Bronze and the Silver views only.



Slack · Sideflix Airflow: sideflix_daily.check_sources FAILED, run alert_test_20261103, attempt 3
Recorded alert of 4 Nov (slack_8_4_alert_received.png). Live, the missing-day run before the talk sends the same kind of alert, with its own run ID.

Evidence: evidence/06_incidents/ (incident_report.md, live_rehearsal_20261105/) · evidence/05_orchestration/ · docs/live_demo_runbook.md

Speakers: Rayan (limits), then each member

What we claim, and what we do not

Limits, stated upfront

- Fictional company: Netflix is only the real-world reference
- Viewing sessions are simulated, and labelled as such
- Sources cover different periods; Sicario is the only confirmed Netflix Top 10 match
- GCS holds only the Terraform state; Snowflake runs on AWS; Airflow and PostgreSQL run locally
- Incidents were injected on purpose, then actually executed
- The catalogue watermark does not see deletions (prototype)
- Shared Snowflake trial account; the course database is never modified

Matis Douay

SOURCES & DATA

- Five sources and manifest, v2 ingestion
- dbt Silver, Gold and tests
- Both incidents, the video, the Snowsight captures

Jérémy Caillaud

PLATFORM & OPERATIONS

- Snowflake account and foundation
- Terraform and its remote GCS state
- Docker/Airflow runs, Slack alerting, roles, costs, first slides

Rayan Faher Bonne

CONSUMPTION

- Owner of the Gold consumption layer
- Presents the Snowsight reporting (read-only analyst role)
- Saved captures executed by Matis

Evidence: README.md, "Scope, limitations and contributions" · docs/architecture.md · evidence/README.md · commit e11c528 (first slides)

Thank you

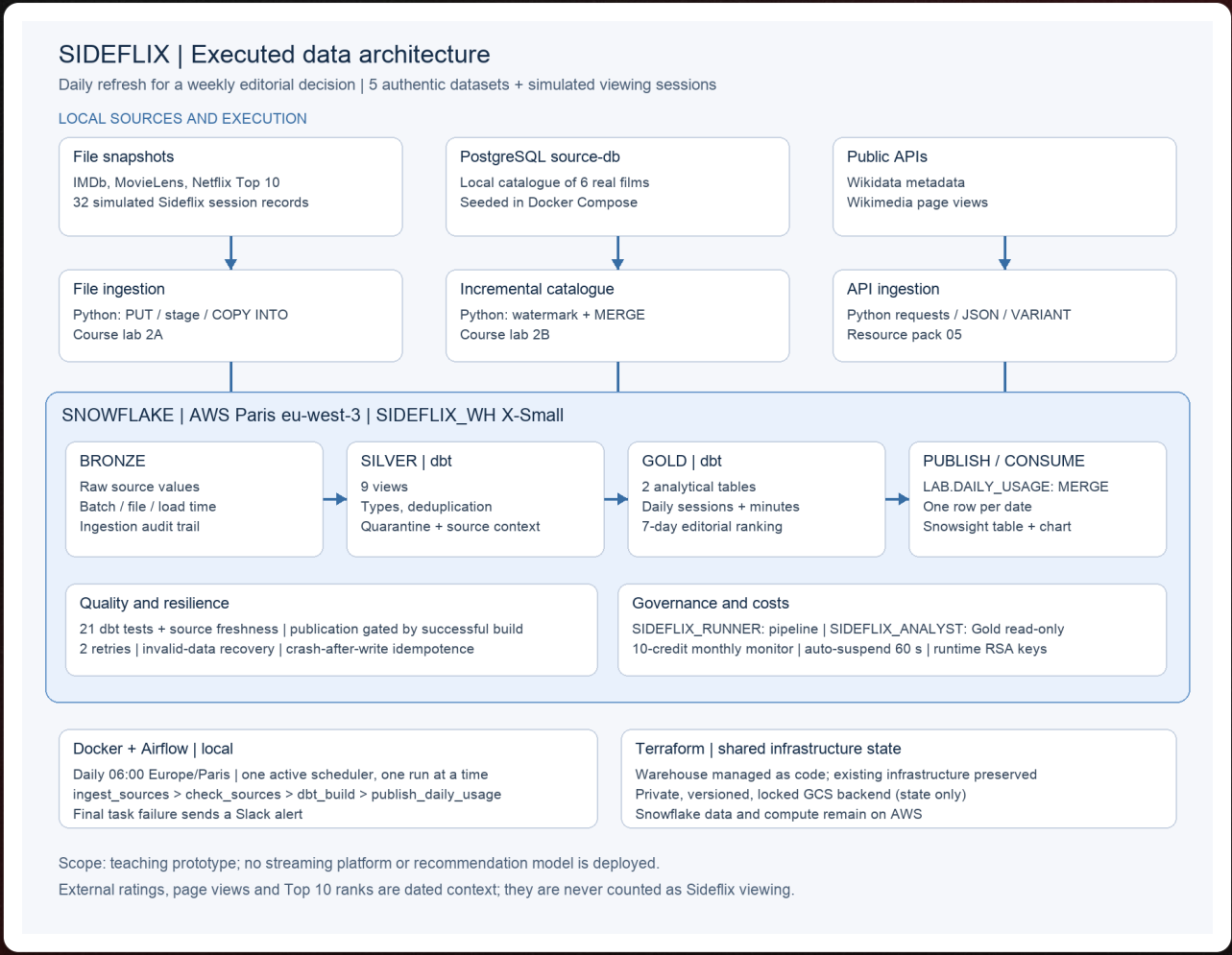
Questions?

13 · 177.4	33 / 33	2	≈ 1.5
sessions · minutes on 1 Nov, one row	dbt operations green	incidents recovered	credits a month projected, cap 10

Repository: github.com/CLDprog/SideFlix-Project
Start here: README.md · evidence/README.md (C1–C8 map) · docs/dbt/index.html

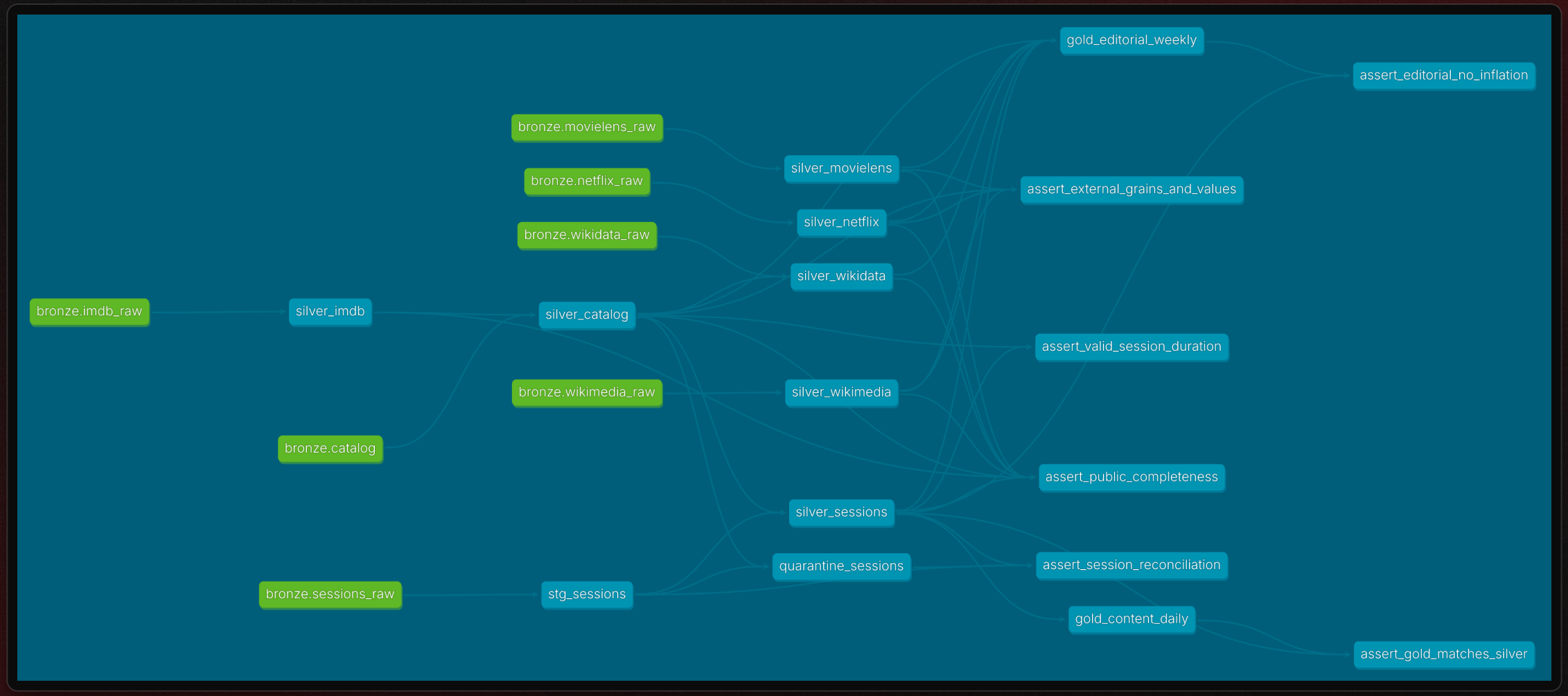
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Architecture diagram, as delivered



Source: docs/architecture.png · details docs/architecture.md · slide 10 is a simplified redraw

dbt lineage, as generated



Source: docs/dbt_lineage.png (dbt docs) · interactive version docs/dbt/index.html · slide 13 is drawn from docs/dbt/manifest.json